Nika Ghaderi

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Education

- **B.Sc. in Computer Engineering, Sharif University of Technology, Tehran, Iran** *2022 – Expected 2027*
  GPA: 19.30/20
- **Farzanegan High School (SAMPAD), Iran** *2022*
  Mathematics and Physics track, total average: 19.96/20

Research Experience

- **Undergraduate Researcher (B.Sc. Thesis), DAISY Lab – Supervisor: Dr. Maryam Ramezani**
  Sharif University of Technology, Tehran, Iran *March 2026 – Present*
  Learning Query Cost Models that Extrapolate Across Data Scale
  - Predicting the latency and I/O of Spark SQL queries from their compile-time query plans, to support cost-based optimization and resource planning.
  - Designing graph-transformer models over query-plan graphs; in a 10× scale-up pilot on real Spark traces, the models achieved ~37% lower latency error than the best template-level analytic cost model.
- **AI Research Assistant, Online Learning Project**
  Dr. Rohban's AI & Robotics Lab, Sharif University of Technology *January – March 2025*
  - Wrote a technical survey of online learning methods (SGD variants, multi-armed bandits, online k-means, online meta-learning), with applications to precision agriculture and computer vision.

Professional Experience

- **C/C++ Software Engineer**
  Sharif HPC Center, Sharif University of Technology *February 2026 – Present*
  GPU-accelerated packet processing for real-time analysis of high-throughput VoIP traffic.
  - Designed a CUDA multi-pattern matching engine for SIP/RTP traffic detection, replacing a production Intel Hyperscan CPU implementation: 100 Gbps end-to-end vs. 10–20 Gbps on an 8-thread CPU baseline.
  - Built lock-free data paths for line-rate ingestion: a ring buffer supporting out-of-order packet placement under burst traffic, and a multi-threaded C++17 pipeline for call-detail-record correlation (zero per-message heap allocation, Kafka integration, deterministic replay).
  - Developed a heuristic protocol-layer scan for tunneled traffic, reducing the false-positive rate from ~25% to zero on observed traces.
- **Software Engineering Intern, OpenAirInterface 5G Project**
  EURECOM, Sophia Antipolis, France *July – September 2025*
  - Accelerated 5G NR channel simulations on the GPU, offloading multipath convolution and noise generation to CUDA: over 400× speedup vs. a single-threaded scalar C baseline.
  - Optimized kernels (shared-memory tiling, cuRAND); benchmarked three CUDA memory models: unified memory trailed, while explicit copy and ATS each led in different configurations.
  - Built a CI-integrated CTest benchmark suite for numerical correctness and regression testing.

Technical Skills

- **Languages:** C, C++17, CUDA, Python, SQL (PostgreSQL), Java
- **Systems & Tools:** Linux (namespaces, cgroups v2), Git, CMake/CTest, Nsight Compute, cuda-gdb, cuRAND, Kafka (librdkafka), Apache Spark (Spark SQL), LaTeX
- **ML & Data:** PyTorch, PyTorch Lightning, DGL, scikit-learn, Optuna, pandas/NumPy
- **Hardware & Low-Level:** Verilog, Quartus, MIPS and x86 (8086) assembly, Ghidra

Teaching and Academic Service

- **Teaching Assistant, Sharif University of Technology** ***2023 – Present***
  Designed and contributed to homework, quizzes, and exams for: Linear Algebra ***(Fa26, Sp26, Fa25, Sp25, Fa23, Sp23)*** – Artificial Intelligence ***(Fa26, Sp26, Fa25)*** – Computer Networks ***(Fa26)*** – Compiler Design ***(Fa26, Sp26)*** – Scientific and Technical Presentation ***(Sp26, Fa25)*** – Logic Circuits ***(Sp26, Sp25, Fa24)*** – Database Design ***(Sp26, Sp25)*** – Formal Languages and Automata ***(Sp26, Sp25)*** – Game Theory ***(Sp26)*** – Computer Architecture ***(Fa25)*** – Fundamentals of Programming in C ***(Fa24)***
- **Grader, Iranian National Olympiad in Artificial Intelligence, Young Scholars Club (YSC)** ***2026***
  Graded exams that determined national medalists in Iran's selection for the International Olympiad in AI (IOAI).

Projects

- Unsupervised Anomaly Detection on MVTec AD ***2026***
  Course project (2 members). Implemented PaDiM and STFPM in PyTorch for defect detection and localization, trained only on defect-free images; PaDiM reached 0.93 macro image AUROC across 15 categories.
- Linux Container Runtime Development ***2025***
  Built a lightweight, daemonless container runtime from scratch in C: filesystem isolation via chroot, process and mount isolation via Linux namespaces (PID, mount), CPU and memory limits enforced with cgroups v2, and a custom CLI with an automated setup script.
- 64B/66B to 8B/10B Encoding Converter with Error Correction ***2025***
  Team project (3 members). Built a real-time Arduino-based converter from 64B/66B to 8B/10B encoding, with Hamming-code error detection/correction and simulated fault injection for resilience testing. – *Demo*
- Strategy Game (Stronghold Crusader–inspired) ***2023***
  Team project (3 members) in Java/JavaFX with an MVC architecture. Implemented A* pathfinding for units, a tile-based map with building, resource, and trade systems, JSON persistence and JUnit tests. – *Demo*
- Vim-Inspired Text Editor ***2022***
  Implemented a command-line text editor in C with insert/cut/copy/paste, find and replace, grep, undo, auto-indent, file comparison (diff), directory tree view, and piping of command output between commands.

Academic Programs and Training

- **Winter University in Engineering Science, SPbPU, St. Petersburg, Russia** ***2025***
  Innovator track; completed program modules in *Artificial Intelligence and Digital Technologies.* Fully funded; selected to represent *Sharif University of Technology* under a bilateral inter-university agreement.
- **Neural Networks and Deep Learning, Coursera (DeepLearning.AI)** – *Credential* ***2025***
- **Front-End Training Course, Quera College – Perfect Score** ***2023***

Honors and Awards

- Ranked 78th of 145,747 candidates nationally (top 0.06%) and 23rd of 52,009 in Region 2, Mathematics track, Iranian Nationwide University Entrance Exam (Concours). ***2022***
- Awarded a gold medal in the 11th-grade *Tehran Math House* competition, Iran. ***2020***
- Placed among Iran's top 5 in the Cayley Contest, organized by *CEMC, University of Waterloo, Canada*. ***2020***
- Awarded the Top Scholar plaque by the *Association of Young Mathematicians*, Iran. ***2017***

Foreign Languages

- Persian: Native | English: IELTS Academic 8.5/9.0 (CEFR C2), October 2026 | French: B1 (CEFR)